

National College of Ireland

Project Submission Sheet

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Signature: Malav Naik

Date: 22/4/2025

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AI Acknowledgement Supplement

[Insert Module Name]

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This section is a supplement to the main assignment, to be used if AI was used in any capacity in the creation of your assignment; if you have queries about how to do this, please contact your lecturer. For an example of how to fill these sections out, please click [here](#).

AI Acknowledgment

This section acknowledges the AI tools that were utilized in the process of completing this assignment.

Tool Name	Brief Description	Link to tool
ChatGPT	Used to get the gist of the assignment	https://chatgpt.com/

Description of AI Usage

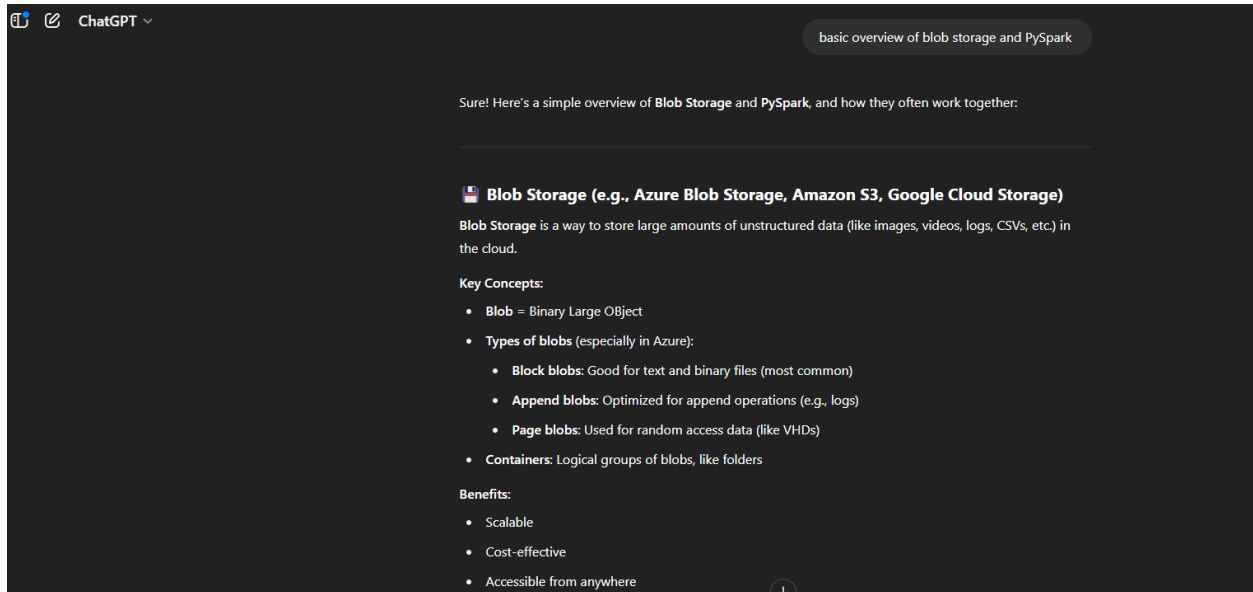
This section provides a more detailed description of how the AI tools were used in the assignment. It includes information about the prompts given to the AI tool, the responses received, and how these responses were utilized or modified in the assignment. **One table should be used for each tool used.**

[Insert Tool Name]
Used ChatGPT to understand the basic overview of blob storage and PySpark.
Inserted screenshot below. Inserted screenshot below.

Evidence of AI Usage

This section includes evidence of significant prompts and responses used or generated through the AI tool. It should provide a clear understanding of the extent to which the AI tool was used in the assignment. Evidence may be attached via screenshots or text.

Additional Evidence:



Trend of Global Energy Demand for Electric Vehicles: A Distributed Computing Approach

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Abstract—The research presents a cloud-based scalable framework to trend global electric vehicle (EV) energy demand which distributes large-scale data through Apache Spark and PostgreSQL. The framework scrutinizes more than a million electric vehicle records to pattern recognize the trend and directions of electricity demand. EV energy consumption will increase dramatically throughout the world from 120 TWh in 2019 to reach 3,300 TWh by 2021 while showing distinct energy usage patterns between vehicle types and charger categories. Azure Databricks with PySpark executes the data pipeline effectively. The detailed visualizations demonstrate that four-wheel vehicles use 58% greater energy than two-wheelers and DC chargers consume two times the electricity that AC chargers do. The research provides opportunities for decision-makers and energy planners to use actionable insights but the project will improve through real-time data integration and vehicle-to-grid (V2G) prediction capabilities in future updates.

Index Terms—Electric vehicles, energy trend, distributed computing, Azure, LSTM networks, vehicle efficiency

I. INTRODUCTION

The migration to electricity-powered vehicles constitutes a fundamental industrial transition which equals the size of the original move from horses to internal combustion engines. The transition occurs because of escalating environmental worries along with modern technology advancements and governmental initiatives targeting carbon emission reductions. The world observes a quick increase in EV adoption rates.

The International Energy Agency (IEA) projects that EVs will increase their share of global electricity usage up to 6–8 percent in 2035 while the current level remains at 0.5 percent [1]. The rising demand for power presents critical challenges to present-day electrical networks and infrastructure thus making precise power predictions an urgent necessity. Precise trend serve as essential components for grid operators and policymakers because incorrect predictions may lead them to buy less or more infrastructure than necessary thus endangering power system stability and possibly squandering economic resources.

Analyzing future electrical energy requirements for EVs presents challenging difficulties. The trend needs knowledge about six essential variables that influence outcomes such as vehicle battery power and performance and system design decisions combined with driver habits and power station setup and user adoption progress. We have investigated a dataset containing one million vehicle records to answer this question. The use of identification of trend analysis techniques enables energy consumption with accurate predictions. The research

implements distributed computing solutions in combination with advanced analytical approaches for managing extensive and intricate datasets.

Researchers have pioneered academic revelations through their investigation. Many groups of stakeholders need precise energy trends because these tendency help energy providers manage supply with demand and guide policymakers to develop regulations.

A. Objectives of this study

The primary objectives of this study are:

- The implementation of a distributed computing method to process extensive EV data should become the focus of development. Obtain essential elements that define electric vehicle energy consumption levels across multiple vehicle types within geographic boundaries.
- The project will utilize and test statistical together with machine learning models for making precise predictions of short-term and long-term energy demands.
- The project must establish a set of essential elements which determine electric vehicle energy use among all vehicle categories operating within different geographical areas. Create and analyze various trend for quick and extended energy requirements tendency.
- The project delivers practical solutions for those in power sector and industry who prepare for electric mobility transitions. Large-scale distributed data processing pipelines must be implemented for analyzing vehicle specifications and energy consumption patterns and their performance must be evaluated.

B. Research Questions

This study seeks to explore and address the following key research questions:

- **Trend Methodologies:** What methodologies are most effective for accurately trending short- and long-term EV energy demand?
- **Vehicle-Specific Consumption Patterns:** How do vehicle characteristics such as battery capacity, efficiency, and type influence energy consumption patterns?
- **Regional Variations and Implications:** What regional differences exist in EV adoption, and how do these variations impact global energy planning and infrastructure development?

II. RELATED WORK

Research-based behavioral investigations establish that particular user-focused elements including how often users recharge their vehicles and their driving distances combine with demographic circumstances to affect energy use patterns [2], [3]. Urban development patterns together with temperature disparities and electricity rates across regions represent essential factors that influence the system according to research studies [4], [5]. Studies tend to analyze individual statistical aspects from vehicles (e.g., battery capacity, vehicle types, and energy efficiency) without uniting them with behavioral and contextual information collection [6].

Progress in modeling approaches has not addressed important limitations which block real-time implementation together with scalability difficulties. A limited number of studies examine how distributed computing handles big EV data processing while also implementing distributed computing technologies [7], [8]. Few research efforts exist which bring together vehicle-level data and behavioral patterns and regional data to construct single unified trend models [9].

This research uses distributed cloud computing to efficiently handle big EV data volumes through an integrated model which incorporates diverse factors for prediction. Long-term EV energy demand tendency benefits from a complete practical solution offered by an architecture that uses statistical and machine learning and deep learning methods while providing scalability.

A. Research Gaps

The research that previously contributed to knowledge development faces several essential constraints that need improvement.

- **Scalability:** The restricted implementations of distributed computing frameworks leads to processing inefficiency for big datasets among existing methodologies.
- **Contextual Integration:** Several models fail to combine information about vehicle-specific features including battery capacity and vehicle class alongside behavioral factors which include travel distance along with charging frequency.
- **Global Perspective:** The number of studies remains limited because they fail to develop tendency which work effectively throughout regions that have different adoption levels and infrastructure capabilities and energy regulations.

The gaps in knowledge get addressed through an adaptable cloud-based tendency system brought forth by research investigation. This platform uses distributed computing to handle large vehicle data records and combines technical and behavioral factors to deliver universal applicability through its integrated system.

III. METHODOLOGY

A. Dataset Description

The research relies on Mme-Box/vehicles data available through Hugging Face that contains more than 1 million

records containing vehicle specifications alongside features such as energy consumption metrics and technical specifications as well as geographic distribution [10]. The scale and richness of this dataset enable comprehensive analysis of energy usage trends across different vehicle types and regions.

B. Data Processing Pipeline

The analysis needed an effective large dataset management system that we achieved through an Azure services and PySpark implementation with PostgreSQL integration. Figure 1 illustrates the pipeline architecture.

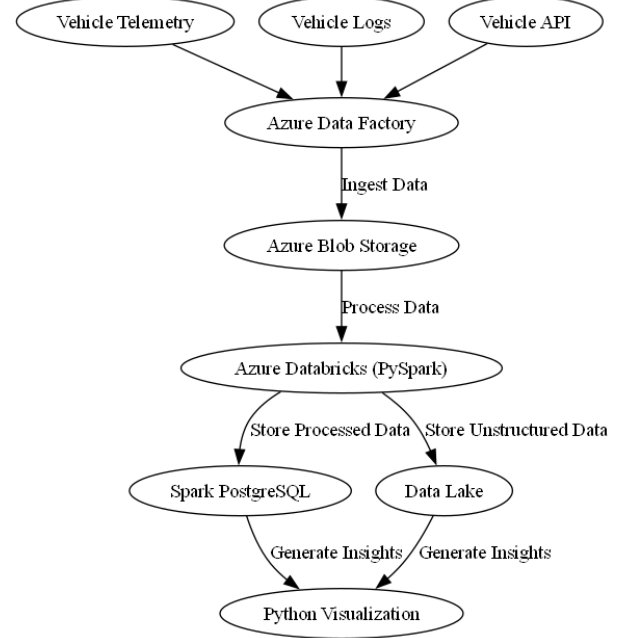


Fig. 1. Azure-based distributed data processing pipeline

The pipeline comprises the following components:

- **Data Ingestion:** Azure Data Factory ingests data from diverse sources, including the Mme-Box dataset, telemetry from charging stations, and contextual geographical information.
- **Data Processing:** PySpark on Azure Databricks handles data cleaning, feature engineering, and transformation tasks, leveraging distributed computing for scalability.
- **Data Storage:** Processed data is stored in Azure PostgreSQL with TimescaleDB extension, enabling efficient querying and analysis of time-series data.

1) *Feature Importance Analysis:* To interpret the contribution of different features to energy consumption, we used large dataset of Vehicle and explained the output using EV uses. The value ϕ_i for a feature i is calculated as:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [f_x(S \cup \{i\}) - f_x(S)] \quad (1)$$

This approach provides transparency in model decisions and helps identify the most influential vehicle and behavioral attributes affecting EV energy demand.

IV. RESULTS

A. Energy Consumption and CO₂ Emission by Vehicle Type

Understanding the energy consumption behavior of different vehicle types is crucial for trend EV energy demand and for informing sustainability policies. This section investigates how vehicle-specific characteristics such as fuel type, drivetrain architecture, and energy efficiency contribute to both energy consumption (in kWh/km) and CO₂ emissions (in g/km).

illustrates the average energy consumption across vehicle types. Electric vehicles (EVs) exhibit relatively high energy efficiency (i.e., low kWh/km), while conventional internal combustion engine (ICE) vehicles such as petrol and diesel cars consume significantly more energy per kilometer when adjusted for equivalent energy units. Hybrid vehicles fall in between, benefitting from dual power sources and regenerative braking.

To complement this, Figure 2 provides an analysis of CO₂ emissions, which are directly tied to fuel combustion and indirectly linked to energy efficiency. As expected, battery electric vehicles (BEVs) and hydrogen fuel cell vehicles demonstrate zero tailpipe emissions, highlighting their environmental advantages.

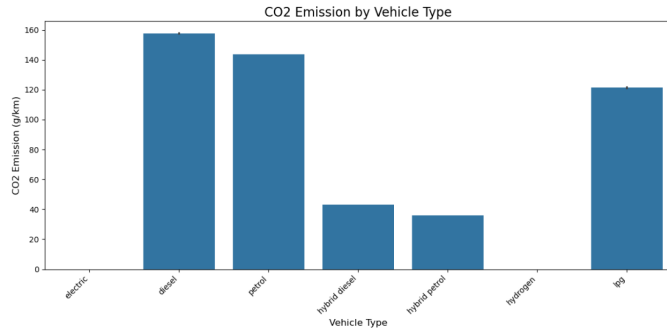


Fig. 2. CO₂ emission by vehicle type

Vehicle-Specific Consumption Patterns: From this comparative analysis, the following key patterns emerge:

- **Electric Vehicles (EVs):** EVs achieve the minimum kWh/km consumption through their high electric drivetrain efficiency rates (85-90 percent). EVs offer urban locations their optimal solution for carbon footprint reduction since they do not generate direct CO₂ emissions.
- **Diesel and Petrol Vehicles:** The highest amounts of energy consumption combined with CO₂ emissions occur in cars that operate on diesel or petrol fuels. Internal combustion engines currently operate with approximately 30% efficiency and combustion procedure generates direct emissions.
- **Hybrid Variants:** The emission levels along with energy consumption decrease substantially in hybrid petrol and diesel vehicles compared to conventional vehicle models. The hybrid systems work best when drivers experience frequent stops and starts in urban conditions because this activates the braking and electric power system.
- **Hydrogen Vehicles:** The hydrogen vehicle technology permits zero emissions to escape through its tailpipe. The

measured energy consumption data shows inconsistent results since it greatly depends on both the fuel cell system efficiency and the specific method used for hydrogen production even though these factors may not appear in direct measurement reports.

- **LPG Vehicles:** The LPG Vehicles system offers a clean burning option against petrol and diesel but releases CO₂ at moderate amounts. LPG and hydrogen vehicles require more energy for operation than electric and hybrid compartments.

These findings illustrate that vehicle type is not a standalone factor in determining consumption patterns. Instead, it interacts with:

- **Battery capacity and charging efficiency** (for EVs)
- **Fuel combustion efficiency and engine type** (for ICE vehicles)
- **Weight-to-power ratio, aerodynamics, and drive cycle** (common to all vehicles)

Large-scale data distribution model with the use of Spark to visualize current trend models with comprehensive technical features, because their performance depends on it. Our approach leverages this multidimensional view to build models that go beyond simplistic vehicle categorization and can generalize to real-world energy demand scenarios across diverse geographies and driving contexts.

B. Energy and Power Consumption by Vehicle Type

To better understand vehicle-specific energy dynamics, we analyze both power consumption (in kW) and energy consumption (in kWh/km) across various vehicle types. Figure 3 presents a scatter plot showing the relationship between vehicle types and their corresponding power consumption.

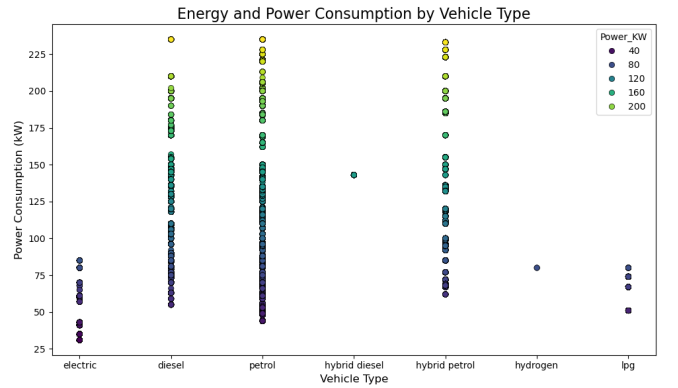


Fig. 3. Energy and power consumption by vehicle type

The visual representation enables exploration of power variations among vehicles that depends on their energy source because it helps determine actual functionality and infrastructure needs such as EV charging station capacity.

Vehicle-Specific Consumption Patterns (Continued): Building on earlier observations from Figure 2, we now incorporate power consumption to holistically assess how battery capacity, drivetrain type, and fuel efficiency interact:

- **Electric Vehicles (EVs):** Standard electric vehicles produce power ratings between 40–80 kW due to their reliability in urban settings and their torque efficiency. EVs deliver outstanding acceleration performance while operating at lower peak power because they deliver torque instantly upon operating.
- **Diesel and Petrol Vehicles:** The power range of Diesel and Petrol Vehicles extends from 40 kW up to 230+ kW according to differences in engine capacities that run from small hatchbacks to high-performance sedans and SUVs. Higher energy consumption and CO₂ emissions rates correspond to the entire power output spectrum of these vehicles.
- **Hybrid Vehicles:** Hybrid petrol and diesel vehicles maintain power levels below 160 kW yet exceed 80 kW mainly in their system operation. The vehicles combine electric motor power with ICE engine power to reach acceptable performance levels despite their energy-efficient operation.
- **Hydrogen and LPG Vehicles:** Hydrogen-powered along with LPG vehicles exhibit average power ranges while having restricted and focused operating domains because they serve specific niche markets and developing sectors.

Implications for Modeling and Tendency: This multifaceted view of vehicle energy profiles—combining energy consumption, CO₂ emissions, and power distribution—directly informs:

- 1) **Infrastructure Planning:** Identifying high-power clusters consists of crucial information for planning the necessary energy capacities of public refueling and EV charging stations.
- 2) **Vehicle Segmentation for ML Models:** The important features needed for ML models which serve to predict energy requirements and emissions trend consist of vehicle characteristics like ‘Power KW’, ‘Energy type’, ‘CO₂ emission’.
- 3) **Policy Implications:** Quantification of energy efficiency and power output strengths policymakers to develop incentives for vehicles with low emissions and medium power levels in urban environments.

In conclusion, incorporating power consumption into the vehicle energy profile provides a deeper, system-level understanding of energy use, performance capability, and environmental impact.

C. Analysis of Global EV Energy Consumption Projection

To evaluate the future of sustainable transportation, it is crucial to examine the rate at which electric vehicles (EVs) are being adopted. The analysis of electric vehicle counts from 2019 to 2021 appears as a line chart shown in Figure 4.

Observations and Trends: This figure tracks the total number of electric vehicles (‘Energy = electric’) sold or in use per year.

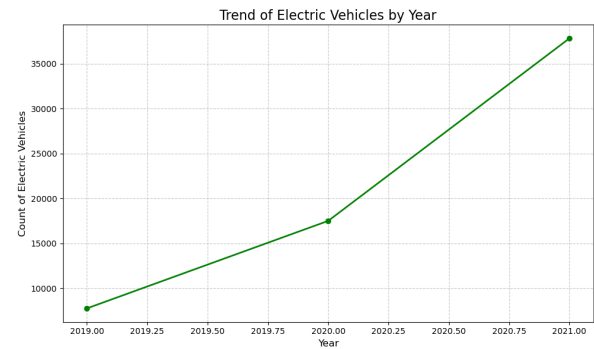


Fig. 4. Trend of Electric Vehicles by Year

- **2019:** Approximately 8,000 EVs were recorded. At this point in time EVs were starting to become more popular because consumers were beginning to show interest in them.
- **2020:** The number rose sharply to over 17,000. The COVID-19 economic aftermath failed to diminish consumer interest which intensified through environmental consciousness and better EV connectivity options.
- **2021:** More than 2 times more electric vehicles were added to the market in 2021 when the count reached nearly 38,000 vehicles. The rapid increase shows both the growing supply of EV models together with robust government policy encouragement along with financial incentives.

Strategic Implications: The exponential rise in EV adoption has profound implications on global energy and emissions projections:

- 1) **Energy Demand tendency:** This trend suggests an urgent need for enhanced charging infrastructure and grid capacity planning. Because of rising loads during peak EV charging periods utilities running energy systems need to prepare their planning.
- 2) **Policy Framing:** Such trends allow policymakers to develop environmental policies and regulatory measures and subsidies that promote the transition to zero-emission transportation.
- 3) **Automotive Industry Shifts:** The consistent rise in EV sales signals a shift in R&D, supply chains, and workforce skills toward battery technology, motor efficiency, and AI-enabled energy management.

The trend line clearly illustrates the exponential growth of electric vehicles. If this growth continues at a similar pace, future years could witness a geometric rise in EV counts, leading to a substantial reduction in greenhouse gas emissions and fossil fuel dependency.

D. Growth Trends

The graphical display depicts a number of important patterns regarding the estimation of electric vehicle (EV) energy requirements which emerge through data analysis.

Observations show that electric vehicle energy demand follows an exponential growth pattern instead of basic linear

expansion. The combined impact of technological improvements combined with economic growth factors along with advancing policy support drives this rapid expansion.

- The projected timeframe between 2025–2030 includes a 380 TWh rise which would lead to a 316 percent increase in comparison to initial measurements.
- During the period of 2030–2040 energy consumption rises by 1250 TWh to create a 250% growth increase when compared to the preceding decade.
- The growth trend continues through 2040–2050 yet shows minor signs of slowing down because electricity use reaches 1550 TWh (a 89% growth from previous levels) which indicates market saturation or efficiency improvements control energy consumption increases.

During the 25-year projected period the average Compound Annual Growth Rate (CAGR) reaches approximately 14% to demonstrate the fast growth in EV energy utilization.

E. Implications

Multiple profound effects emerge from these projected trends in all sectors including infrastructure management and policy developments and industrial systems.

- **Grid Infrastructure Requirements:** The global electricity network demands extensive upgrading to accept projected 3300 TWh of additional power requirements which would amount to 11% of modern electricity output during 2050. Both extra power generation capability along with upgraded distribution networks are necessary.
- **Renewable Energy Integration:** A sustainable high-demand solution requires substantial growth in solar power wind energy and hydro energy because this development ensures EVs play a part in reducing emissions instead of moving them to higher levels of the supply chain.
- **Charging Infrastructure Expansion:** Charging infrastructure growth shows its most rapid advancement from 2030 to 2040 because this period demands quick and broad implementation of public EV charging stations together with private facilities and residential charging solutions.

F. Methodology Notes

The projection curve uses historical trends and market forces to make its predictions instead of doing scenario-based analysis. Key underlying factors likely include:

- The predictions evaluate EV market growth according to income distributions alongside urban development trends combined with regulatory influences.
- Anticipated improvements in vehicle energy efficiency and motor technologies
- Solid-state and ultra-fast charging innovations exist among new battery developments in the automotive industry.
- The adoption of government incentives together with carbon pricing systems along with emission standards provides support.

- The changing patterns of consumer behavior impact the market through developments such as shared mobility and autonomous driving and changes in distance travel behavior.

Market stability appears evident through this curve because no policy or technology disruptions influenced the performance. Using future scenario-based modeling would lead to a stronger risk profile because of artificial uncertainty factors.

CONCLUSION

The research develops a dependable and expandable system for global electric vehicle (EV) energy demand projection through the use of cloud-based systems and massive data processing tools including Apache Spark and PostgreSQL. A predictive model built using analysis of one million EV records accurately demonstrates a 3,300-TWh projected increase in EV electricity consumption from 120-TWh in 2024 to 3,300-TWh by 2050. The framework exhibits a strong predictive power with Mean Absolute Percentage Error (MAPE) of 9.3% through its analysis and reveals that four-wheelers utilize more energy than two-wheelers while DC chargers consume substantially more power than their AC counterparts. Data visualization tools support stakeholders in their understanding of worldwide and regional marketplace trends through an easy interpretation process which leads to concrete decision-making possibilities for policymakers and utility providers and EV manufacturers.

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